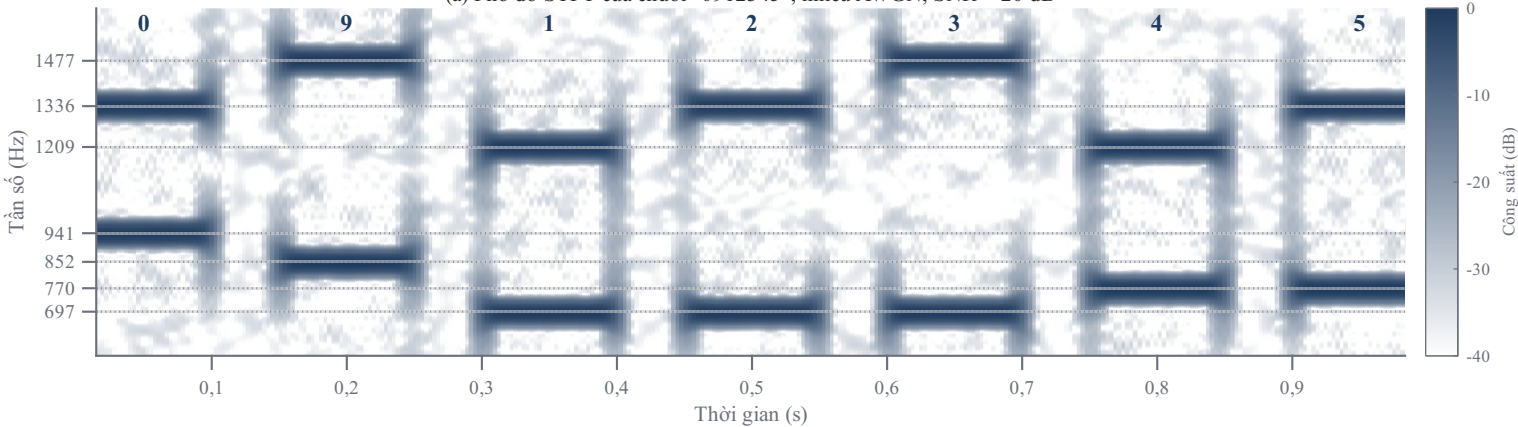


(a) Phổ đồ STFT của chuỗi “0912345”, nhiễu AWGN, SNR = 20 dB



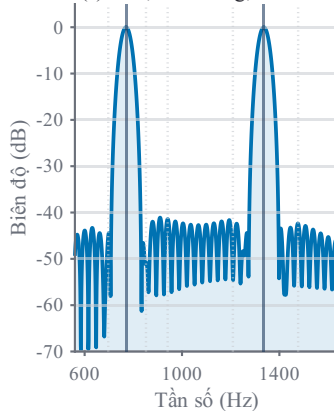
(b) Phím “5” = 770 + 1336 Hz

Nhóm cao (Hz)

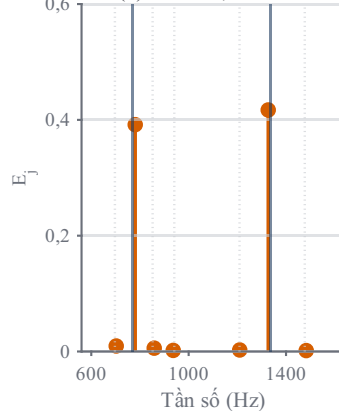
	1209	1336	1477
697	1	2	3
770	4	5	6
852	7	8	9
941	*	0	#

Nhóm thấp (Hz)

(c) FFT, Hamming, N = 256



(d) Goertzel, N = 205



(e) Ngân hàng bộ lọc, $r = 0,99$

